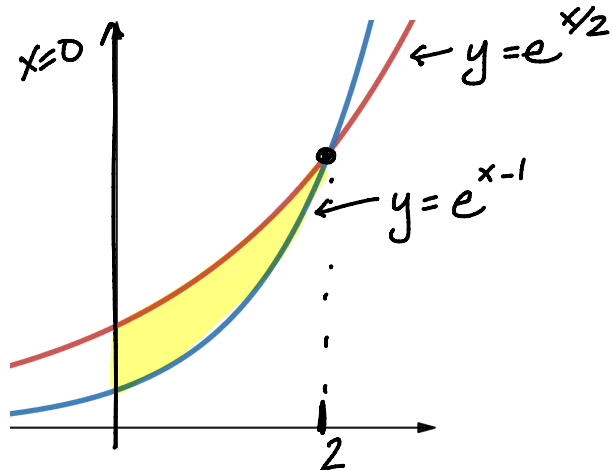


1. (a) (3 pts) Use the graphs below to shade the region bounded by $y = e^{x/2}$, $y = e^{x-1}$ and $x = 0$.



find pt. of intersection:

$$e^{x/2} = e^{x-1} \text{ if}$$

$$\frac{x}{2} = x - 1$$

$$x = 2x - 2$$

$$2 = x$$

- (b) (7 pts) Determine the area of this region using an appropriate integral.

$$A = \int_0^2 (e^{x/2} - e^{x-1}) dx = \left[2e^{x/2} - e^{x-1} \right]_0^2$$

$$= (2e^{2/2} - e^{2-1}) - (2e^{0/2} - e^{0-1})$$

$$= 2e - e - 2 + e^{-1}$$

$$= e + \frac{1}{e} - 2$$

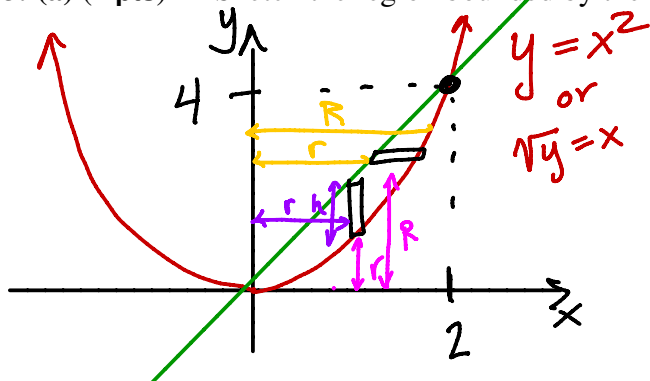
2. (5 pts) Set up but do not evaluate an integral computing the arc length of the curve $y = \tan(x^2)$ between $x = -\pi/4$ to $x = \pi/4$.

$$L = \int_{-\pi/4}^{\pi/4} \sqrt{1 + (f'(x))^2} dx = \int_{-\pi/4}^{\pi/4} \sqrt{1 + 4x^2 \sec^4(x^2)} dx$$

$$y = \tan(x^2)$$

$$y' = \sec^2(x^2)(2x) = 2x \sec^2(x^2). \text{ So } (y')^2 = 4x^2 \sec^4(x^2)$$

3. (a) (2 pts) Sketch the region bounded by the curves $y = x^2$ and $y = 2x$.



pt of intersection:

$$x^2 = 2x$$

$$x^2 - 2x = 0$$

$$x(x-2) = 0$$

$$x = 0, x = 2$$

- (b) (6 pts) Use an integral to compute the volume of the solid found by rotating the region in part a. around the x -axis.

$$V = \pi \int_a^b (R^2 - r^2) dx = \pi \int_0^2 ((2x)^2 - (x^2)^2) dx$$

$$= \pi \int_0^2 (4x^2 - x^4) dx = \pi \left(\frac{4}{3} x^3 - \frac{1}{5} x^5 \right)_0^2 = \pi \left(\frac{32}{3} - \frac{32}{5} \right)$$

$$= \pi \left(\frac{2 \cdot 32}{15} \right) = \frac{64\pi}{15}$$

- (c) (4 pts) Use the **shell method** to set up an integral to calculate the volume of the solid obtained by rotating the region in part a. around the y -axis. You do not need to evaluate the integral.

$$V = 2\pi \int_a^b r h dx = 2\pi \int_0^2 x \cdot (2x - x^2) dx = 2\pi \int_0^2 (2x^2 - x^3) dx$$

- (d) (4 pts) Use the **slicing method** (disks/washers) to set up an integral to calculate the volume of the solid obtained by rotating the region in part a. around the y -axis. You do not need to evaluate the integral.

$$V = \pi \int_a^b (R^2 - r^2) dy = \pi \int_0^4 (\sqrt{y})^2 - \left(\frac{y}{2}\right)^2 dy = \pi \int_0^4 \left(y - \frac{1}{4}y^2\right) dy$$

4. (10 pts) A 3-meter long whip antenna has linear density $\rho(x) = 5 - \frac{1}{x+1}$ grams per centimeter (starting at $x = 0$). Determine the mass of the antenna. Include units.

$$\text{mass} = \int_a^b \rho(x) dx = \int_0^{300} \left(5 - \frac{1}{x+1}\right) dx = 5x - \ln(x+1) \Big|_0^{300}$$

$$3 \text{ meters} = 300 \text{ cm.} \quad = (5 \cdot 300 - \ln(300+1)) - (5 \cdot 0 - \ln(0+1))$$

$$= 1500 - \ln(301) \text{ grams}$$

5. (10 pts) A 1-meter spring requires 20 J to compress the spring to a length of 0.9 meters. How much work would it take to compress the spring from 1 meter to 0.8 meters?

① Find k .

$$W = 20 \text{ J} = \int_0^{0.1} kx dx = \left. \frac{k}{2} x^2 \right|_0^{0.1} = \frac{k}{200}$$

$$\text{So } 4000 = k.$$

② Find W .

$$W = \int_0^{0.2} 4000x dx = 2000x^2 \Big|_0^{0.2} = \frac{2000}{25} = 80 \text{ J}$$

6. Evaluate the definite integrals. Simplify your answers

(a) (7 pts) $\int_0^{\pi/4} \tan \theta \, d\theta$

$$= \int_0^{\pi/4} \frac{\sin \theta}{\cos \theta} \, d\theta = -\ln(\cos \theta) \Big|_0^{\pi/4} = -\ln(\cos(\pi/4)) + \ln(\cos(0))$$

$$= -\ln(\sqrt{2}/2) + \ln(1)$$

$$= \underline{-\ln(\sqrt{2}/2)} = \underline{\underline{\ln(\sqrt{2})}}$$

(b) (7 pts) $\int_0^2 x e^{3x} \, dx$

IBP
 $u = x \quad dv = e^{3x} \, dx$
 $du = dx \quad v = \frac{1}{3} e^{3x}$

$$= x \cdot \frac{1}{3} e^{3x} \Big|_0^2 - \frac{1}{3} \int_0^2 e^{3x} \, dx$$

$$= \left(\frac{2}{3} e^6 - 0 \right) - \left[\frac{1}{9} e^{3x} \right]_0^2$$

$$= \frac{2}{3} e^6 - \left(\frac{1}{9} e^6 - \frac{1}{9} e^0 \right) = \boxed{\frac{5}{9} e^6 + \frac{1}{9}}$$

7. Evaluate the indefinite integrals.

(b) (6 pts) $\int \sin^3(4x) \cos^2(4x) dx$

$$= \int \sin^2(4x) \cos^2(4x) \sin(4x) dx$$

$$= \int (1 - \cos^2(4x)) \cos^2(4x) (\sin(4x) dx)$$

$$\begin{aligned} \text{let } u &= \cos(4x) \\ du &= -4 \sin(4x) dx \\ -\frac{1}{4} du &= \sin(4x) dx \end{aligned}$$

$$= -\frac{1}{4} \int (1 - u^2) u^2 du = -\frac{1}{4} \int (u^2 - u^4) du = -\frac{1}{4} \left(\frac{1}{3} u^3 - \frac{1}{5} u^5 \right) + C$$

$$-\frac{1}{12} \cos^3(4x) + \frac{1}{20} \cos^5(4x) + C$$

(a) (6 pts) $\int \sec^4(x) dx$

$$= \int \sec^2 x (\sec^2 x dx)$$

$$= \int (1 + \tan^2 x) \sec^2 x dx$$

$$\begin{aligned} \text{let } u &= \tan x \\ du &= \sec^2 x dx \end{aligned}$$

$$= \int (1 + u^2) du = u + \frac{1}{3} u^3 + C$$

$$= \tan(x) + \frac{1}{3} \tan^3(x) + C$$

(c) (6 pts) $\int \arcsin(x) dx$

IBP

$$u = \arcsin(x) \quad dv = dx$$

$$du = \frac{1}{\sqrt{1-x^2}} dx \quad v = x$$

$$= x \arcsin(x) - \int \frac{x dx}{\sqrt{1-x^2}}$$

$$\text{let } u = 1-x^2, \quad du = -2x dx$$

$$\frac{1}{2} du = -x dx$$

$$= x \arcsin(x) + \frac{1}{2} \int u^{-1/2} du$$

$$= x \arcsin(x) + u^{\frac{1}{2}} + C$$

$$= x \arcsin(x) + \sqrt{1-x^2} + C$$

(c) (6 pts) $\int \frac{2}{(2x+1)(2x-3)} dx$

partial fractions

$$\frac{2}{(2x+1)(2x-3)} = \frac{A}{2x+1} + \frac{B}{2x-3} \quad \text{or}$$

$$2 = A(2x-3) + B(2x+1)$$

$$\text{if } x = -\frac{1}{2}: 2 = A(-1-3) = -4A$$

$$A = -\frac{1}{2}$$

$$\text{if } x = \frac{3}{2}: 2 = B(2(\frac{3}{2})+1) = B(4)$$

$$B = \frac{1}{2}$$

$$= -\frac{1}{4} \ln|2x+1| + \frac{1}{4} \ln|2x-3| + C$$

$$= \ln \left(\left| \frac{2x-3}{2x+1} \right|^{\frac{1}{4}} \right) + C$$

8. (10 pts) Use the method of Trigonometric Substitution to evaluate the integral $\int \frac{dx}{(4+x^2)^2}$. Your final answer must be simplified and written in terms of x .

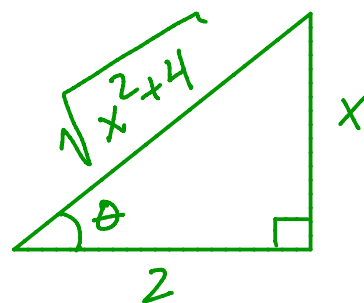
$$\begin{aligned} \text{Let } x &= 2 \tan \theta \\ dx &= 2 \sec^2 \theta d\theta \\ 4+x^2 &= 4+4\tan^2 \theta \\ &= 4\sec^2 \theta \end{aligned}$$

$$\begin{aligned} \int \frac{dx}{(4+x^2)^2} \\ = \int \frac{2 \sec^2 \theta d\theta}{16 \sec^4 \theta} = \frac{1}{8} \int \frac{d\theta}{\sec^2 \theta} \end{aligned}$$

$$= \frac{1}{8} \int \cos^2 \theta d\theta = \frac{1}{16} \int (1 + \cos(2\theta)) d\theta = \frac{1}{16} \left(\theta + \frac{1}{2} \sin(2\theta) \right)$$

$$= \frac{1}{16} (\theta + \sin \theta \cos \theta)$$

$$= \frac{1}{16} \left(\arctan\left(\frac{x}{2}\right) + \frac{2x}{x^2+4} \right)$$



$$\frac{x}{2} = \tan \theta$$

$$\sin \theta = \frac{x}{\sqrt{x^2+4}}$$

$$\cos \theta = \frac{2}{\sqrt{x^2+4}}$$

Extra Credit. A particle moving along a straight line has a velocity of $v(t) = te^{-t}$ after t seconds where v is measured in meters per second.

(a) (2 pts) How far does the particle travel from time $t = 0$ seconds to time $t = T$ seconds?

$$\begin{aligned} \text{distance travelled} &= \int_0^T t e^{-t} dt & u=t \quad dv=e^{-t} dt \\ & & du=dt \quad v=-e^{-t} \\ &= -te^{-t} \Big|_0^T + \int_0^T e^{-t} dt = -Te^{-T} - \left[e^{-t} \right]_0^T = -Te^{-T} - (e^{-T} - e^0) \\ &= -e^{-T}(T+1) + 1 \text{ meters.} \end{aligned}$$

(b) (3 pts) Use your answer from part a. to determine how far the particle travels in the long-term, as $T \rightarrow \infty$.

$$\lim_{T \rightarrow \infty} \left(-e^{-T}(T+1) + 1 \right) = \lim_{T \rightarrow \infty} \left(1 - \frac{T+1}{e^T} \right) = 1$$

The particle only moves 1 meter.

You may find the following **trigonometric formulas** useful. Other formulas, not listed here, should be in your memory, or you can derive them from the ones here.

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

$$\sin(ax) \sin(bx) = \frac{1}{2} \cos((a-b)x) - \frac{1}{2} \cos((a+b)x)$$

$$\sin(ax) \cos(bx) = \frac{1}{2} \sin((a-b)x) + \frac{1}{2} \sin((a+b)x)$$

$$\cos(ax) \cos(bx) = \frac{1}{2} \cos((a-b)x) + \frac{1}{2} \cos((a+b)x)$$